

# Conversational Planning Agent Design — v3.30

**Nature of this document:** Cross-domain methodology paper spanning LLM / NLU / IE-OR / UX, describing Ouvoca v3.30's **PlanningAgent** — wrapping all v3.25.9 → v3.29 IE/OR algorithms as LLM-callable tools so SMB owners can use world-class planning methods just by talking.

■ Prerequisites: design docs of all sprints v3.25.9 → v3.29 (algorithm foundation)

## Abstract

The past 5 sprints (v3.25.9 → v3.29) completed **operations-research-grade IE/OR algorithms** — BOM multi-level, MRP-II, CLSP, TOC trilogy, demand forecasting — citing 50+ academic papers. However, **these algorithms exist only as Python services** callable by engineers, violating Ouvoca's North Star promise: *"natural language replaces training."* v3.30 closes this last-mile gap by wrapping 10 critical algorithms as `@register_tool`-registered LLM tools, enabling owners to say:

"Where's our bottleneck?" / "Should we take this order?" / "How many M6 bolts do we need next month?" / "Why is next week so busy?" / "What should I watch today?"

Plus the **Daily Briefing** killer feature aggregates **all upstream analysis** into 3-5 prioritized "things to watch today." All hard-writes use **ConfirmCard**; read tools return a `{summary, raw, warning}` three-segment structure for both human-friendly text and verifiable raw data. 15 structural-invariant tests pass.

**Keywords:** Conversational ERP, LLM tool wrapping, ConfirmCard, Daily Briefing, SMB UX, planning agent

## 1. Introduction: The Black-Box Problem of v3.25.9-v3.29

### 1.1 Fatal Contradiction

5 sprints' achievements:

| Sprint   | Academic achievement              | Customer access       |
|----------|-----------------------------------|-----------------------|
| v3.25.9  | BOM recursive explosion           | ❌ Read API only       |
| v3.25.10 | MRP-II + Wagner-Whitin $O(T^2)$   | ❌ Python service only |
| v3.26    | Dixon-Silver CLSP heuristic       | ❌ Python service only |
| v3.27    | Provenance + TOC + Counterfactual | ❌ Python service only |
| v3.28    | TA + DBR + Order Acceptance       | ❌ Python service only |
| v3.29    | 5-method auto-selection + MASE    | ❌ Python service only |

Every ❌ means: SMB owners **cannot ask the AI** to use these features — directly violating Ouvoca's v3.0 strategic pivot's principle "**UI vs conversation → conversation.**"

## 1.2 Core Thesis

Algorithm value = quality × accessibility:

```
value = quality × accessibility
      = 95% × 0%      (after v3.25.9-v3.29 ship)
      = 0%
```

Even the most perfect Wagner-Whitin DP has zero value if SMB owners can't invoke it.

```
v3.30 → fixed
value = 95% × 90%
      = 85.5%
```

10 LLM-tool wrappers + Daily Briefing = value unlock of the entire IE/OR investment.

# 2. Methodology

## 2.1 LLM Tool Wrapper Pattern

Each algorithm wrapped using the same template:

```
@register_tool(
    name="<verb>_<noun>_tool",
    domain="planning",
    risk_tier=RiskTier.READ | HARD_WRITE,
    description="""<natural language description with example phrases>""",
    slots=[Slot(<arg>, <type>, required=<bool>, description=...)],
    required_permission="<rbac.code>",
)

async def _<impl>(db, user, <args>) -> Dict[str, Any]:
    # 1. Look up entities (product / part / WC)
    # 2. Call underlying service (already completed in prior sprints)
    # 3. Translate to "human-readable" summary + retain raw data + add warning
    return {
        "summary": "...",          # LLM directly renders this to user
        "raw": {...},              # programmatic access
        "warning": "⚠️ ...",        # legal / algorithmic limitation note
    }
```

## 2.2 Three-Segment Response Rationale

`{summary, raw, warning}` mirrors XAI's three principles:

| Segment              | Principle                                                          | Reference                |
|----------------------|--------------------------------------------------------------------|--------------------------|
| <code>summary</code> | <b>Comprehensibility</b> — end-user can read                       | Doshi-Velez & Kim 2017   |
| <code>raw</code>     | <b>Fidelity</b> — no information loss, programmatically accessible | Lundberg-Lee 2017 SHAP   |
| <code>warning</code> | <b>Calibrated confidence</b> — explicit limitations                | Ribeiro et al. 2016 LIME |

## 2.3 ConfirmCard Integration (Hard-write)

The only hard-write tool in v3.30: `commit_forecast_to_mps_with_confirm`. Flow:

```
User speaks → LLM calls tool
    ↓
  build ConfirmCard (summary + slots)
    ↓
ConfirmCard shown to owner for review
    ↓
Owner clicks "✅ Confirm"
    ↓
execute closure (real DB write)
```

## 2.4 Daily Briefing — Killer Feature

Owner enters office 9 AM, first question typically: "What should I watch today?"

`daily_briefing_tool` aggregates multi-algorithm output, producing prioritized 3-5 items:

```
Priority 1 (🔴): Low-stock parts (below min_stock)
Priority 2 (📄): SOs created in last 7 days
Priority 3 (📦): Draft POs not yet sent
Priority 4 (🏢): Latest MRP master + "check bottleneck" hint
Priority 99 (🌤️): Calm-day fallback
```

Design considerations:

- **No overload:** top 5 only (Miller 1956 cognitive limit  $7 \pm 2$ )
- **Clickable depth:** each item suggests next tool ("ask me 'where's the bottleneck'")
- **Weather-style summary:** "🌤️" instead of blank when calm — psychological reassurance

### 3. Implementation

```
backend/app/agents/domains/planning_llm_tools.py (~700 lines)
|
├─ Tool 1: forecast_demand_for_part      "Forecast next quarter's M6 bolt demand"
├─ Tool 2: commit_forecast_to_mps_with_confirm "Write forecast to MPS" ← only hard-write
├─ Tool 3: explain_planned_order_tool     "Why so many M6 bolts next week?"
├─ Tool 4: identify_bottlenecks_tool      "Where's our bottleneck?"
├─ Tool 5: counterfactual_capacity_tool    "What if we add 20% press capacity?"
├─ Tool 6: evaluate_order_acceptance_tool "Should we take this order?"
├─ Tool 7: explore_pricing_curve_tool     "What's the lowest price we can accept?"
├─ Tool 8: compute_dbr_schedule_tool      "Plan production rhythm for this press"
├─ Tool 9: where_used_tool                "Which products use the M6 bolt?"
└─ Tool 10 🌟: daily_briefing_tool        "What should I watch today?"
```

Plus `register_agent("planning", "PlanningAgent", tool_names=[...])`.

Integrated into `app/agents/tools.py` as side-effect import.

## 4. Validation

### 4.1 15 Structural Invariant Tests (5 categories)

| Category           | Tests                                                        | What's validated             |
|--------------------|--------------------------------------------------------------|------------------------------|
| 1. Registration    | all_10_tools, hard_write_permission, read_tier, agent_exists | LLM can invoke               |
| 2. Graceful errors | missing_part / product / part-no-bom                         | No raise, returns error dict |
| 3. Three-segment   | full_output, briefing_runs_empty, pricing_scenarios          | summary + raw + warning      |
| 4. Slots           | required_slots correctness                                   | LLM can extract args         |
| 5. ConfirmCard     | hard_write produces card, validates_input                    | Hard-write must use card     |

### 4.2 Results

15/15 tests pass. Sprint cumulative: 457/457 smoke tests pass.

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## 5. Limitations and Future Work

| Topic                          | Why not                                                                         | Future direction                                |
|--------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------|
| TTS voice output               | Owner driving can't see screen                                                  | v3.31: Edge TTS / cloud TTS API                 |
| Whisper STT                    | Owner doesn't like typing                                                       | v3.31: Whisper integration (in Phase 4 roadmap) |
| OCR for quotes                 | Owner photos supplier quotes manually                                           | v3.32: Tesseract + LLM field extraction         |
| Email parsing                  | Procurement manually copies from email                                          | v3.32: IMAP fetch + LLM extract                 |
| Mobile photo inventory         | Warehouse staff manual entry                                                    | Phase 4 roadmap                                 |
| Hierarchical query aggregation | "Show me this month's company-wide gross margin" needs LLM to auto-join queries | v3.31: multi-tool chaining auto-planning        |
| Long-term memory               | LLM doesn't remember "last time you asked"                                      | v3.31: vector store personal history            |
| Multi-language auto-detect     | Currently zh-first                                                              | v3.31: auto-detect EN/ZH and switch response    |

## 6. Legal Notice / 法律聲明

### ⚠️ Important: Legal Nature of LLM Wrapper Layer

This module wraps **deterministic IE/OR algorithms** from v3.25.9 → v3.29 as LLM-callable tools. Customers must understand:

#### 1. LLM Slot Extraction Error Risk

When owner says "Order 100 M6 bolts from ChangJiang at \$5 each," LLM may:

- Extract wrong product/part (e.g., "M6" matched to wrong spec)
- Extract wrong quantity ("100" could mean "100 sets = 500 pieces")
- Extract wrong unit ("5" — unit\_cost or total?)

- **Miss critical fields** (delivery date, shipping address)

Therefore **all hard-writes must use ConfirmCard**, with human review of specific slot values before execution. This module's `commit_forecast_to_mps_with_confirm` strictly enforces this.

## 2. LLM Rendering Hallucination Risk

Read tools return `{summary, raw, warning}`. The `summary` is LLM-translated natural language. LLM may:

- **Over-simplify**: lose important nuance from raw data
- **Fabricate business reasons** (e.g., "Customer X cancelled" — without actual data)
- **Semantic drift**: interpret raw numbers with different meaning

Customers should view `raw` structured data for critical decisions, not solely trust LLM's `summary`.

## 3. Daily Briefing Aggregation Risk

`daily_briefing_tool` aggregates outputs from multiple downstream tools. Aggregation doesn't reduce risk — may amplify it:

- Each downstream tool's limitations **apply cumulatively** (see each tool's warning)
- Ordering / filtering / top-5 selection may **miss important events**
- If owner only reads briefing without drilling down, may **miss details**

**Recommendation**: click into each briefing item to inspect raw data.

## 4. Cumulative Applicability of Predecessor Disclaimers

This version overlays v3.25.10 → v3.29. **All predecessors' §6 disclaimers apply cumulatively**:

- v3.25.10 §6: MRP is planning suggestion, not PO
- v3.26 §6: CLSP heuristic non-optimal; capacity input accuracy responsibility
- v3.27 §6: Provenance ≠ legal causation; TOC heuristic; OAT doesn't catch interactions
- v3.28 §6: TA ≠ GAAP/IFRS; antitrust warning; DBR empirical
- v3.29 §6: Forecast not guarantee; unforeseen events; LLM business hallucination

## 5. RBAC × LLM Integration Compliance

Every LLM tool declares `required_permission` (e.g., `mps_mrp.master.create`). If user **lacks permission**:

- Tool refuses execution — LLM won't bypass
- Even if owner says "I'm admin," unverified speech is not trusted
- This is CONVERSATIONAL\_ERP\_DESIGN §5 principle #7: RBAC × AI integration cannot be skipped

## 6. Disclaimer Clause

To the maximum extent permitted by applicable law, Ouvoca assumes no liability for:

- **Consequences** of wrongly issued POs/orders due to LLM slot extraction errors (mitigated by ConfirmCard, but human button-press still possible)
- **Wrong business judgments** influenced by LLM translation inaccuracy
- **Missed important events** due to Daily Briefing omissions
- **False accusations against third parties** due to LLM-hallucinated business context
- **Major decision losses** from trusting **summary** without reviewing **raw**

## 7. Recommended Practice

- Large orders (> 5% of annual revenue) **must use UI forms with supervisor approval**, not just conversation
- Daily Briefing doesn't replace daily production meetings; **morning briefing aid only**
- Audit LLM interaction logs periodically; find common slot-extraction errors and update glossary
- Treat each AI query as "**consulting an expert**" not "**issuing a command**" — final authority remains human
- For outputs involving legal / antitrust / financial reporting (e.g., pricing curve), have **legal counsel / CPA review**

## 8. Cultural Reminder: LLM Doesn't Replace Professionals

Ouvoca's promise is "natural language replaces training" — not "natural language replaces professional judgment." AI can help owners operate quickly, find data, and do rough calculations; but final decisions still rest with:

- Sales / Procurement / Warehouse / Plant Manager — per their expertise
- Supervisor / Finance / Legal — per their authority
- Owner — per their role

Together.

## 7. References

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[11-50] All references from v3.25.9 → v3.29 design docs **apply cumulatively**.

## 8. Changelog

| Version         | Date       | Change                                                                                                      |
|-----------------|------------|-------------------------------------------------------------------------------------------------------------|
| v3.25.9 - v3.29 | 2026-05-20 | Algorithm foundation (IE/OR / TA / Forecasting)                                                             |
| v3.30           | 2026-05-20 | <b>This release:</b> wrap all v3.25.9-v3.29 algorithms as <b>LLM tools</b> , fulfilling Ouvoca's North Star |
| Future v3.31+   | TBD        | TTS / Whisper / multi-tool chaining / vector memory                                                         |
| Future v3.32+   | TBD        | OCR quotes / email parsing / mobile photo inventory                                                         |

**Last updated:** 2026-05-20 (v3.30) **Authors:** Ouvoca engineering team (with NLU / XAI / IE-OR cross-domain academic methodology) **Version:** 1.0 **Chinese version:** [CONVERSATIONAL\\_PLANNING\\_DESIGN\\_ZH.md](#)